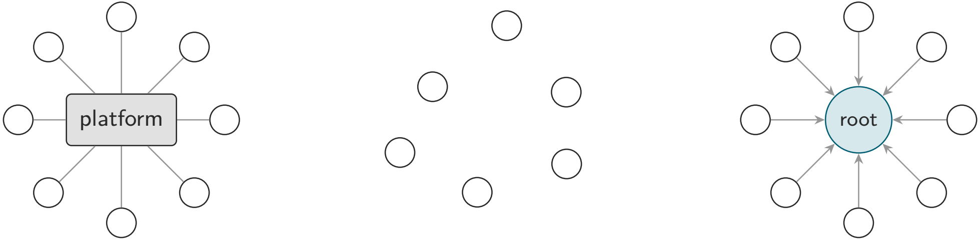
Combinatorial marketplace without a platform

the loop economy on Swarm

Peter Földiák September 2026

loopmarket · ontodag · recordstore · factbond github.com/petfold

Two failure modes, one cause



the platform is the market
rent, gate, ranking

every shop an island
no discovery, no liquidity

the market is a root
anyone recomputes it

- Either the market is somebody's property, or there is no market at all.
- Missing piece: a *shared, unowned* market structure.

The rent, the gate, the ranking

- Take rates: app stores 15–30%; Amazon referral 8–15% of price, plus fulfilment and ads, all-in around half of a seller's revenue. Marketplace Pulse
- Gatekeeping: delisting is unilateral and unappealable.
- Ranking: the platform's algorithm decides who exists.

Platform power is structural: whoever holds the only index can charge for inclusion and shape the market by omission.

The everyday misery: buying one thing for a renovation

1. find which online shops even serve your area
2. search each one
3. work out which products meet your criteria
4. compare prices *with* delivery: some only tell you at the last step
5. compare the shops' conditions, delivery options and dates
6. the one you found does not deliver to your country
7. pick-up would be cheaper: now search for where
8. choose the shop; choose a payment type (some cards carry fees)
9. revise your choice
10. wait, and hope
11. stay at home at the predicted delivery time
12. phone them about why it is late; find the tracking number
13. find the carrier's website; paste the number
14. stay at home at the actual delivery time
15. hope the product meets your criteria after all

And that is a well-defined *good*. Your bathroom tap leaks; the gas heater will not start. Where do you even begin?

It's a mess, and we can do better: **discovery, selection, comparison, logistics, delivery**.

The trades that never happen

traveller, 00:05
wants toothbrush
at hotel reception, $r = 10\,\mathrm{m}$
00:05 – 00:35

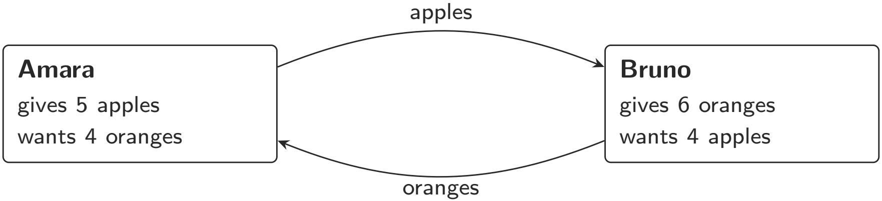
neither knows the other exists

petrol station, 300 m away
gives toothbrush
at the forecourt
open all night

The want is precise: a thing, a place, a time. The give is there. The trade fails on *search*: you would have to go looking, so you don't. By far the most, and the best, potential interactions are lost this way.

You should not have to search. The system should solve it.

The simplest loop: barter

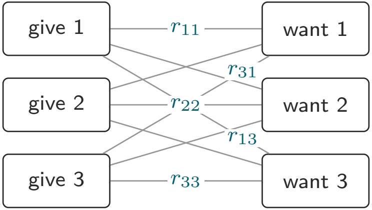


Walk one apple round the loop: Bruno turns it into $6/4 = 1.5$ oranges, Amara turns those back into $1.5 \times 5/4 = 1.875$ apples.

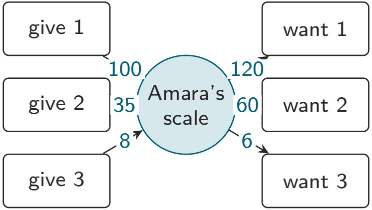
More comes back than went out $\Rightarrow$ the loop closes, with room to spare.

One exchange that satisfies both offers: **4 apples for 5 oranges**.

Private units, for internal bookkeeping only



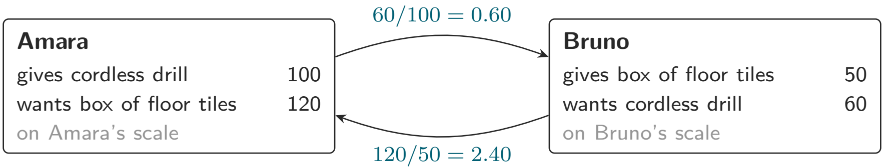
$n \times m$ rates to state
and they can contradict each
other: $r_{11}\,r_{22} \neq r_{12}\,r_{21}$



$n + m$ prices on one private scale
every rate is a ratio p_j/p_i:
no arbitrage possible

The scale is pure bookkeeping: nothing is issued, held or transferred, and each maker's scale is their own. Only ratios ever leave the maker; around a loop the scales cancel.

The simplest loop, on two private scales



One barcode each: the things need no negotiation, only the rate does. Each rate is the taker's bid over the giver's ask, in mixed units. Around the loop the units cancel:

$$\prod \text{rate} = \prod_i \frac{\text{bid}_i}{\text{ask}_i} = \frac{120}{100} \cdot \frac{60}{50} = 1.44$$

Nothing is held or transferred on either scale.

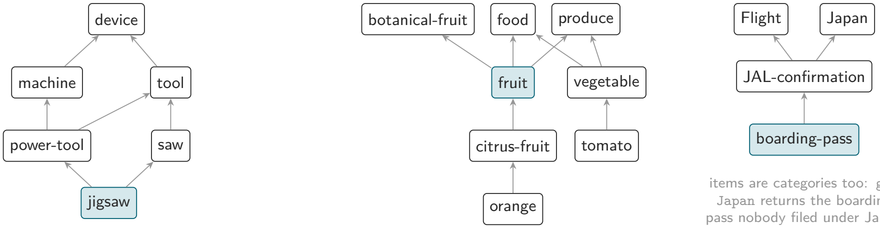
One uniform offer

offer	id = sha256(canonical bytes)
maker	amara (= feed owner)
side	give
thing	{piano-lesson} unit: course
price	100 on Amara's own scale
service	2026-09-01 .. 2027-01-01
region	disc (46.05, 14.50), r = 5 km
valid	2026-09-01 .. 2026-10-01
pins	catalogue root 05ca09... registry 4.1
carried	bond, oracle, arbitrator (enforced in P3)

Every intention has this one form. Exactly one side is a thing, the other is the maker's own scale; a want is the same record with the side flipped.

- A thing, a time, a place, a validity, a price.
- Give or want: the same form.
- Priced on the maker's *own scale*: nothing is held or transferred; the numbers cancel inside a loop.
- A shop, a person, a courier, a bus with empty seats, a stablecoin: all makers.

OntoDAG: categories as a DAG, not a tree



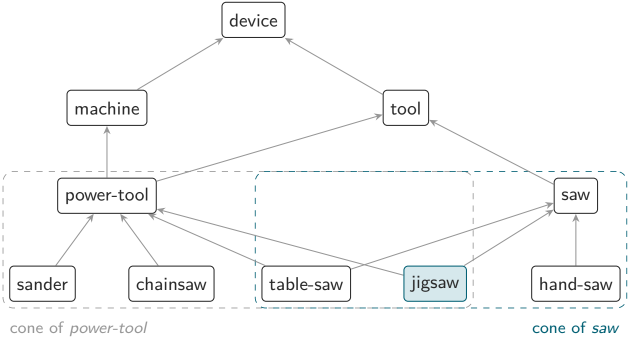
a jigsaw is a saw *and* a power tool

fruit has three parents; membership is reachability

items are categories too: `get Japan` returns the boarding pass nobody filed under Japan

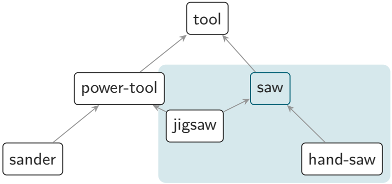
- One relation, *fits within*; one query, the intersection of descendant cones: `get food produce`.
- Typed dimensions are ordinary categories: `time(2026-09-07)`, `weight(..5kg)`, a geohash cell, with the order computed, never stored.
- Canonical form: equal content ⇒ equal root. A catalogue is adoptable by fingerprint and pinned by every offer.

The catalogue: a want is a conjunction, a give is a point



one relation, $\sqsubseteq$ (fits within), always drawn upwards. A want for {saw} is met by any saw, hand or powered; a want for {power-tool, saw} is met by the *intersection* of the two cones: table-saw, jigsaw. The give of a jigsaw satisfies both. Multiple parents are normal: a DAG, not a tree.

ontodag as the language of offers



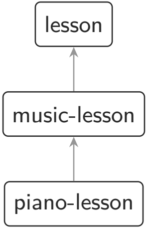
Chen wants a *saw* for the weekend: anything in its cone satisfies her.

Amara gives a *jigsaw*: inside the cone ⇒ match. Bruno's *sander*: outside ⇒ no.

Unknown category ⇒ never matches (fails closed).

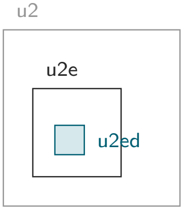
- Match = the give's categories lie inside every cone the want names.
- Shipped this week: the `core` pack, 4,137 consensus categories (WordNet, SUMO, OpenCyc, Wikidata, schema.org as witnesses) + ten domain packs.
- Every offer pins the catalogue root it was written against: a match is reproducible forever.
- Goods are rich (toaster, jeans, jigsaw); services are the next pack.

Fits-within, three ways



meanings

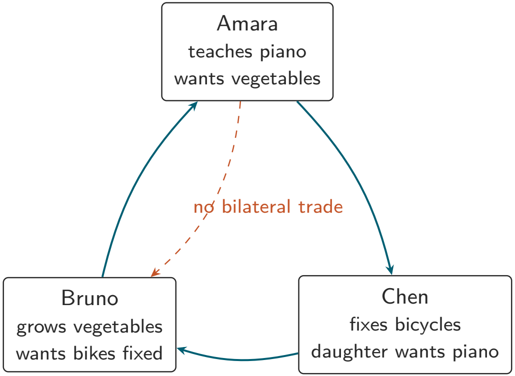
minutes

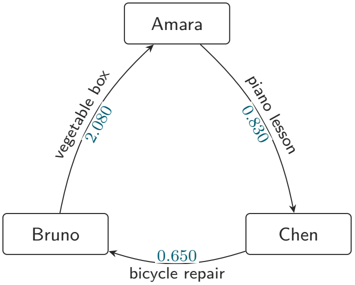


map cells

the same relation everywhere: a piano lesson is a lesson, Tuesday 10–11 is in week 37, cell u2ed is in u2e. The catalogue orders all three by fits-within; exact time and geo checks remain the truth at match time.

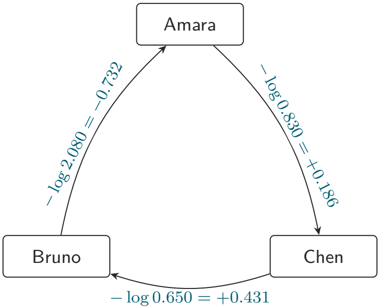
Loops: the trade nobody asked for





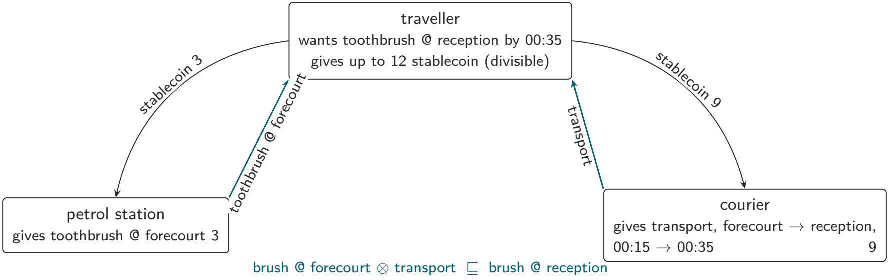
no pair can trade $0.830 \times 0.650 \times 2.080 = 1.122$ surplus 12.2%

A profitable loop is a negative cycle



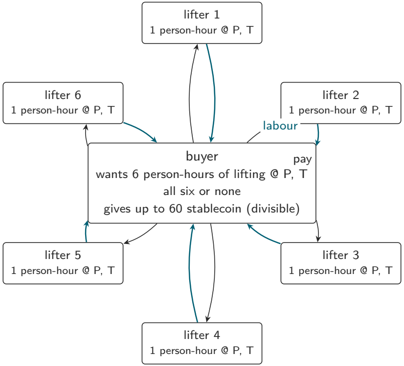
weights $w = -\log(\text{rate})$; $\prod \text{rate} > 1 \iff \sum w < 0$
a profitable loop is a negative cycle; Bellman–Ford finds it in sorted order, deterministically

The toothbrush, solved: one want, many gives



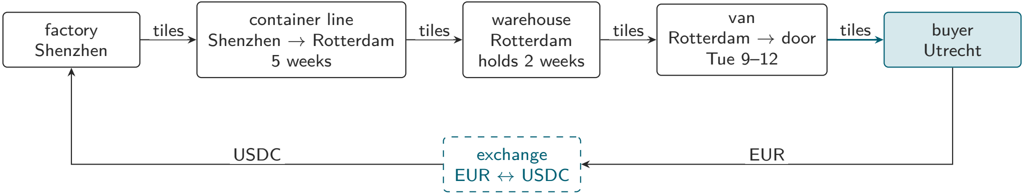
Every give is unconditional and names one plain thing; the courier sells carriage, not toothbrushes. The conjunction lives in the traveller's *one* want, so all-or-nothing is free. The traveller pays once: the money give is divisible and clearing splits it at the per-leg prices. Not a cycle: a balanced flow with a hub. Every node nets to zero on its own scale.

One want, six gives: the buyer pays once



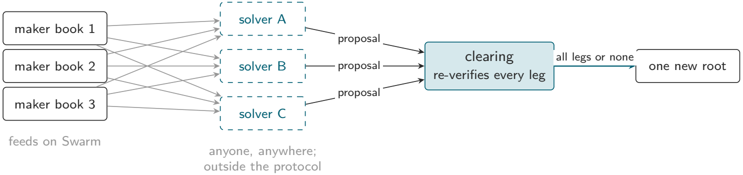
Six unconditional gives of one person-hour each; one want with a minimum quantity of six, so five lifters clear nothing. One divisible money give, split six ways at the clearing prices. Six two-leg trades joined at the hub; a loop is the smallest case of a balanced flow.

Longer loops: place, time and currency are just more legs



Shipping moves a thing in *place*, storage in *time*, an exchange moves value across *currencies*: each is an ordinary maker whose give and want differ in one dimension. The buyer pays in euros, the factory is paid in USDC, and the intermediaries are paid the same way (payment legs not drawn). The solver composes the chain; clearing commits every leg or none.

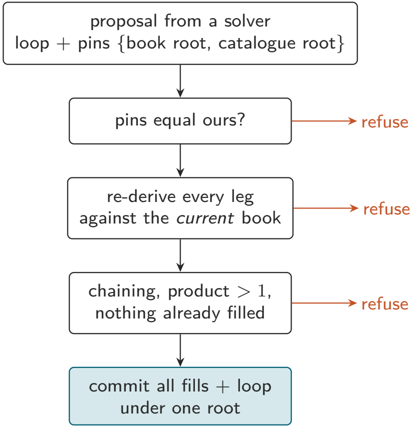
Solvers: outside the protocol, competing, trusted with nothing



A solver reads the pinned books, hunts loops and proposes. It is trusted with nothing: clearing re-derives every leg against its own snapshot of the books and catalogue. The loopmarket repo ships a deterministic baseline solver, the species the others have to beat.

- The CoW Protocol shape: anyone runs a solver; the best proposal wins; the solver is paid from the surplus it found, an endogenous spread bounded by the reserve bid.
- Solvers may be as clever, private and stochastic as they like. The baseline in the repo is deterministic and is the reserve bid.
- P2: sealed proposals per beat, numeraire-free scoring, a fairness floor. docs/plans/P2-batch-auction.md

Clearing trusts no one



the solver is a hint, never an authority. Atomicity is one recordstore commit: either every leg fills or none does.

Who commits? Not the solver

maker
signs the offer once
publishes on own feed
may go offline

offer

solver
reads pinned books
proposes a loop
never writes the book

proposal

clearing
re-verifies every leg
writes `fill/` + `loop/`
one commit, one root

new root

everyone
folds the clearing feed
like any maker's
filled = a record, not a delete

A signed offer is a standing commitment: a match within its terms needs no further consent.

Two proposals for one offer: today first valid wins; P2 packs offer-disjoint winners per beat.

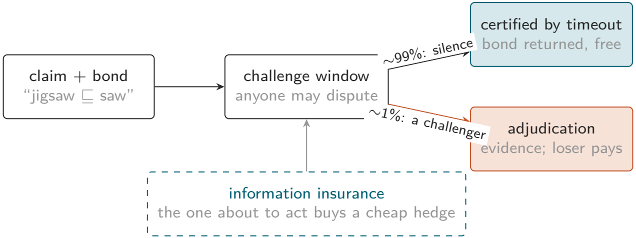
today: one process with its own signed feed. The one point of trust.

P2: a contract on Gnosis. The winning solver submits the transaction; the contract verifies inclusion and absence under the pinned root, then records the fills.

Clearing records the obligations; the goods still have to move. That is *settlement*, off-chain, and it is what P3's bonds, oracles and factbond certificates secure.

- **Clearing** fixes the obligations, instantly and trustlessly. **Settlement** is the makers delivering: slow, physical, and what P3 secures.
- An offer is consumed by a `fill/` record pointing at its loop; the offer itself is never edited or deleted. A stale solver proposes over a filled offer and is simply refused.
- U11: after every fold, each loop has a fill for every leg and each fill points at a present loop.

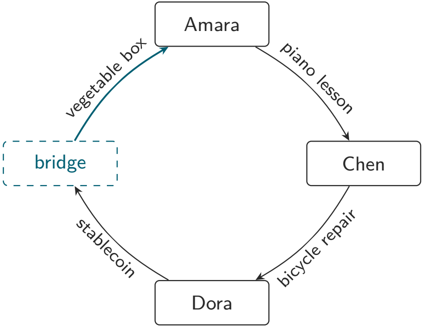
Where verification cannot reach: factbond



Prediction markets cannot scale down to millions of mundane, near-certain facts (matched counterparty, full collateral, mandatory settlement). A bonded assertion needs none of that: undisputed claims certify for free; real price discovery spins up only where someone disagrees. The quoted premium becomes a market-priced reliability signal for every fact.

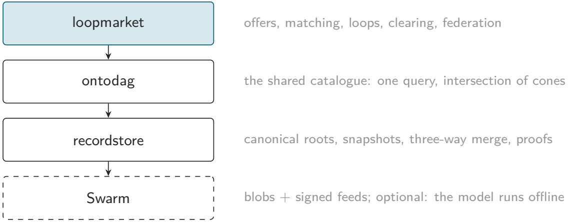
In loopmarket (P3): bonds on the $\sqsubseteq$ edges a loop relied on, on oracle attestations (“delivered at reception, 00:21”), on identity facts; loss experience per edge is the catalogue’s reliability audit. github.com/petfold/factbond, design stage

Money is just another offer



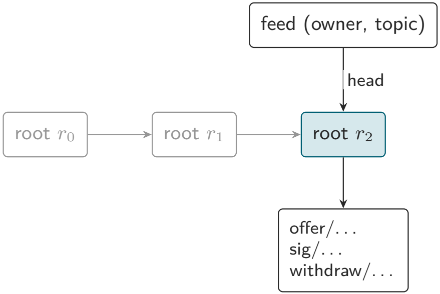
Dora can only pay in money; Amara wants vegetables. A *bridge* is an ordinary maker whose thing is a currency: it buys the vertical's staples for stablecoin and sells them at a spread. Almost-loops failing on a money leg become loops.

The stack, and why it is Swarm-shaped



- Book = versioned keyspace, canonical roots: equal content $\Rightarrow$ equal reference.
- One book per maker, under the maker's own feed and signer. Identity *is* the feed owner.
- Everything so far runs on a *light* node.
- Postage TTL is the offer's lifetime; withdrawal is a tombstone, never a delete.

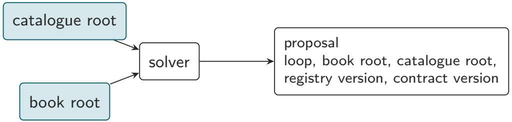
A maker's book is a feed



one maker, one feed, one signer. Identity is the feed owner address. Every version is a content-addressed root; a withdrawal is an added tombstone, never a delete; the postage stamp's TTL is the offer's real lifetime.

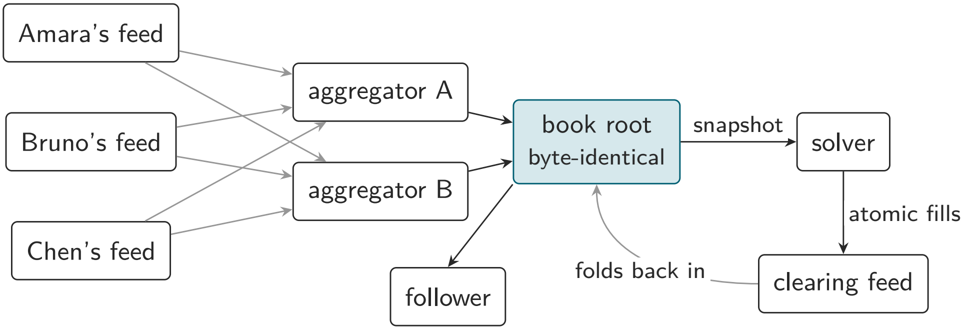
Invariants the tests refuse to let you break

- U1 uniform offer form
- U2 immutable, content-addressed
- U3 clearing trusts no solver
- U4 solve against pinned roots
- U5 positive rates only
- U6 deterministic baseline solver
- U7 vocabulary fails closed
- U11 no partially-filled loop survives a merge



frozen snapshots in, pins out. The same book yields the same loop on every replica, and a verifier a year later can replay the decision. Reproducibility beats freshness.

The federated book



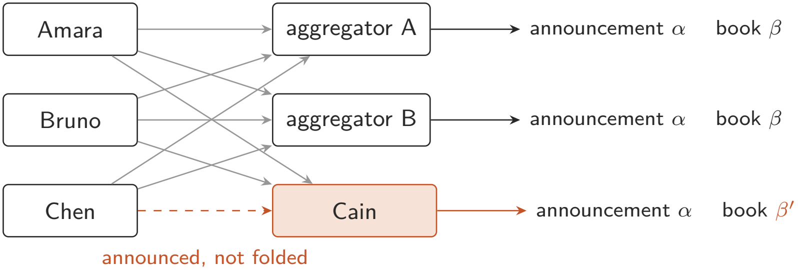
the fold is a pure function of the announced feeds: two aggregators, any order, one root. A reader may use a manifest as a cache or fold the feeds itself.

Demo: three feeds, two honest aggregators, one censor

- Three makers publish under their own feeds; two aggregators fold in different orders: byte-identical manifests.
- Mallory forges an offer in Amara's name: refused at the fold, with an attributed reason.
- Bruno withdraws: a tombstone that survives merges.
- Cain announces Chen's book and silently drops it.
- Clearing bases its own book on the fold; the loop clears; a follower reads six atomic fills from the manifest alone.

`examples/demo_federation.py` – in memory ∼5 s; live on a Bee light node 2–7 min (recording).

Omission is a proof, not a suspicion



same inputs claimed, different fold: evidence by construction.
audit(Cain's manifest) = Chen's two offers, each with an absence proof against β'
that anyone verifies with no store.
a solver folding the announced feeds itself lands on β.

Are aggregators necessary?

- Not for correctness, trust or permission: the fold is pure; any reader recomputes it.
- For read cost: a feed lookup costs seconds; a solver folding 1,000 maker feeds pays 1,000 lookups per beat.
- Direction: solver-side folding is the default; manifests are disposable caches.
- Discovery via a public log (Gnosis registry events as the floor, GSOC as the fast path), never via a party who can omit.

Question for this room: GSOC reception from light nodes, and pub/sub. Content-routed GSOC (one address per catalogue cone $\times$ geo cell $\times$ day) would make the index *the address space*. Stamps are not the problem; delivery guarantees are.

Who pays for what

	pays	earns
maker	postage for their own book: the stamp's TTL *is* the offer's lifetime; no gas, ever	the trade, at or above their asking price
solver	clearing gas for the loops it wins	an endogenous spread, bounded by the reserve bid
aggregator	its own storage and bandwidth	fees for *serving* fast, never for inclusion
clearing	nothing of its own	nothing: gas is the winner's
the protocol	**nothing, in either direction**: no fees, no token, no rewards, no treasury. Every actor bears its own real cost, so a wash loop through sybils can never be paid for by the system it games.	

Ruling 2026-08-21; docs/plans/P2-batch-auction.md §9. Revisit triggers: solver monoculture, aggregator scarcity, measured statistics pollution.

Roadmap

P0	P1	P2	P3	P4
in-memory prototype done	Swarm book, federation live; read path open	verifiable clearing on Gnosis; auctions; pricing	guarantee fabric bonds, oracles (factbond)	privacy commitments, ZK fits-within

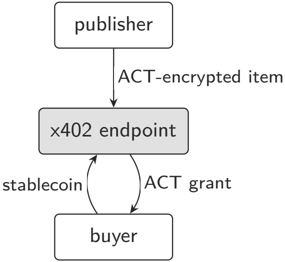
The one point of trust today is clearing. P2 removes it: inclusion and absence under the pinned book root, verified on Gnosis with recordstore's canonical-trie proofs.

Swarm AI Data Exchange and loopmarket: same substrate, same idioms

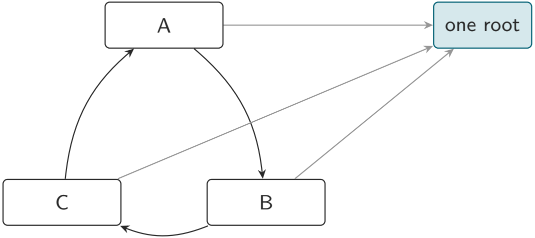
- Both: content-addressed catalogues on Swarm; an atomic root in a feed; feed owner as identity; a light node suffices to read.
- Exchange: Agent Card (ERC-8004) → catalog feed → Mantaray root → `item.jsonld`; ACT-encrypted content; x402 purchase endpoint.
- loopmarket: per-maker feed → recordstore root → offers; aggregator manifests; clearing feed.

What the exchange got right and shipped: ACT per purchase, schema.org-first vocabulary, server minimization as a principle, a working purchase flow.

A storefront and a clearing house



storefront: one seller, one buyer,
one item, one currency, one server



clearing house: k parties, no common currency,
no maker server, one atomic commit

The differences

	Swarm AI Data Exchange	loopmarket
trade shape	bilateral: one publisher, one buyer, one item	multilateral loops; no pair needs to want each other
price	set by the publisher, in a stablecoin	discovered; each maker on their own scale; surplus when $\prod$ rates > 1
medium	required (stablecoin via x402)	none required; money is a bridge offer
described	data assets (schema.org + `swarm-cat:`)	anything: catalogue conjunction $\times$ time $\times$ place
required server	the publisher's x402 endpoint	none for makers; clearing only (on chain in P2)
trust	ERC-8004 reputation + facilitator	re-verification; bonds where verification fails; reputation derived, never a gate
privacy	ACT per purchase	none yet (P4)
maturity	working purchase flow	prototype; live on Swarm; mock clearing

Why not reputation-based

- eBay: 0.3% of transactions rated negative, yet $P(\text{negative} \mid \text{partner rated negative}) > 37\%$: retaliation suppressed truthful feedback until one-sided ratings in 2007. Resnick & Zeckhauser
- Cheap ratings inflate toward uselessness. Filippas, Horton, Golden, EC'18
- loopmarket's rule U12: standing counts cleared, cost-borne loops only; loss experience comes from bonded, adjudicated events (factbond).

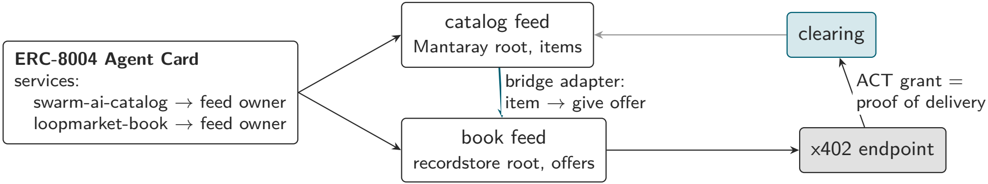
Reputation answers "should I trust this seller". loopmarket makes the question unnecessary where it can, and expensive to game where it cannot. Reputation becomes a statistic you derive, not an institution you believe.

x402 and ERC-8004 fit: as legs, identities and proofs

- **x402** is a bilateral rail; it cannot settle a k-leg loop. It fits three places the plan already has: the stablecoin leg of a bridge offer (settlement cargo, never the scoring numeraire); paid aggregator *serving*, never inclusion; solver fee collection.
- **ERC-8004 identity**: an Agent Card service entry naming the maker's book feed owner is exactly the announcement channel, with the chain registry as the censorship-resistant floor.
- **ERC-8004 reputation**: publish one-sided, aggregated signals derived from cleared loops; consume feedback as one input to risk premia; never gate a trade on it.
- **ERC-8004 validation**: clearing receipts and factbond certificates are validation events with proofs.

x402 for the money leg, loopmarket for the parts x402 cannot see.

How we could converge



one identity, two services; the exchange's catalog becomes book liquidity unchanged; x402 carries the money leg and its ACT grant is the cheapest fulfilment oracle. Kept out of the shared data model: a single currency, a discovery indexer that can omit.

How we could converge

1. One ERC-8004 Agent Card carrying both a `swarm-ai-catalog` and a `loopmarket-book` service.
2. Bridge adapter: every catalog item is a give offer priced in a stablecoin; publish the catalog into a book unchanged.
3. The x402 endpoint as a fulfilment oracle: an ACT grant is a verifiable proof of delivery.
4. Shared vocabulary: the exchange's schema.org/`swarm-cat:` terms as an ontodag pack.
5. Discovery as a public log, not an indexer: keep "no synthesized server views", add "no party who can omit".
6. Keep the one-currency assumption out of the data model: a price is a number on *some* scale.

In return: ACT as a P4 Tier-1 candidate, x402 pragmatism, ERC-8004 identity, and the habit of shipping.

Asks to the room

- GSOC reception from light nodes; pub/sub timing.
- A services layer for ontodag's core pack: the goods are in; repair, tutoring, cleaning are not.
- A first vertical: digital services, agents first.

The root is the market. Swarm holds the books. The chain is the floor for discovery. The one remaining point of trust is clearing, and it is next. github.com/petfold:

loopmarket, ontodag, recordstore, factbond

Backup: a match is an edge

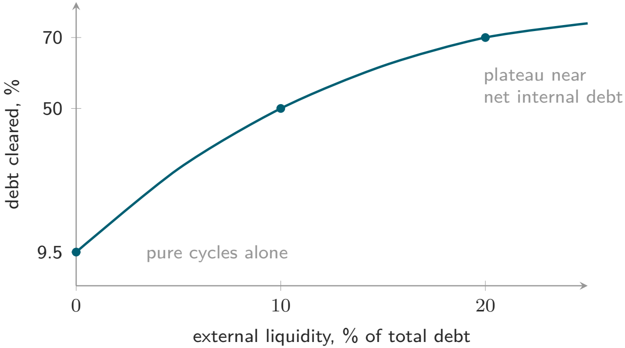
give Amara
{piano-lesson}, 100 on Amara's scale

$\xrightarrow[\text{rate} = 83/100]{\text{check\_match}}$

want Chen
{music-lesson}, 83 on Chen's scale

exact pairwise check: thing fits within (catalogue), windows overlap, discs intersect, both valid now, same pins
$\Rightarrow$ one edge Amara $\to$ Chen in the exchange graph

Backup: a small bridge of liquidity clears most of the rest



1.28M Italian B2B invoices (arXiv:2507.22309, Fig. 10): a small bridge of liquidity turns a cycle-only regime into near-full clearing. Points are read from the paper's figure; the curve is a guide.

Backup: Graphviz renderings (dot2tex)